As an example, consider an oceanographer measuring water depth using a *sonar* system. Short bursts of sound are transmitted from the ship, reflected from the ocean floor, and received at the surface as an echo. Sound waves travel at a relatively constant velocity in water, allowing the depth to be found from the elapsed time between the transmitted and received pulses. As with all empirical measurements, a certain amount of error exists between the measured and true values. This particular measurement could be affected by many factors: random noise in the electronics, waves on the ocean surface, plant growth on the ocean floor, variations in the water temperature causing the sound velocity to change, etc.

To investigate these effects, the oceanographer takes many successive readings at a location known to be *exactly* 1000 meters deep (the true value). These measurements are then arranged as the histogram shown in Fig. 2-11. As would be expected from the Central Limit Theorem, the acquired data are normally distributed. The *mean* occurs at the center of the distribution, and represents the best estimate of the depth based on all of the measured data. The *standard deviation* defines the width of the distribution, describing how much variation occurs between successive measurements.

This situation results in two general types of error that the system can experience. First, the mean may be shifted from the true value. The amount of this shift is called the **accuracy** of the measurement. Second, individual measurements may not agree well with each other, as indicated by the width of the distribution. This is called the **precision** of the measurement, and is expressed by quoting the standard deviation, the signal-to-noise ratio, or the CV.

Consider a measurement that has good accuracy, but poor precision; the histogram is centered over the true value, but is very broad. Although the measurements are correct *as a group*, each individual reading is a poor measure of the true value. This situation is said to have poor *repeatability;* measurements taken in succession don't agree well. Poor precision results from **random errors**. This is the name given to errors that change each

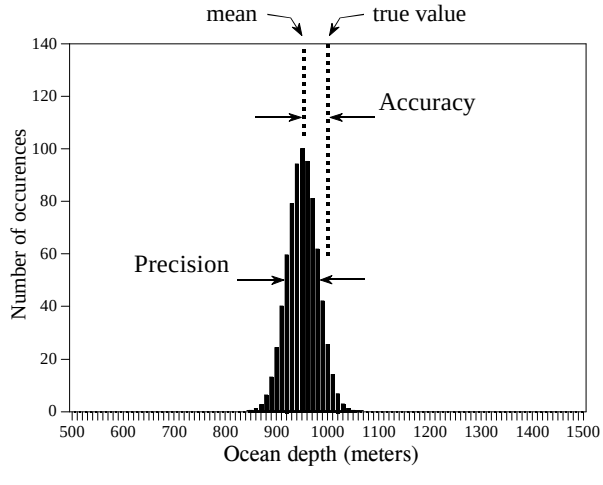

FIGURE 2-11
Definitions of accuracy and precision. Accuracy is the difference between the true value and the mean of the under-lying process that generates the data. Precision is the spread of the values, specified by the standard deviation, the signal-to-noise ratio, or the CV.